Minimum Distance from a Point to a Line



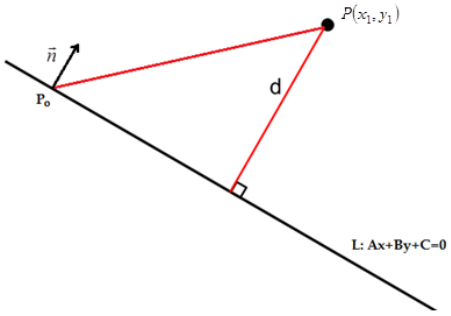


The shortest distance from a point to a line is the _____ distance.

Minimum distance in R^2

Recall: the magnitude of the vector projection of $\vec{a}$ onto $\vec{b}$ is $\dfrac{|\vec{a} \bullet \vec{b}|}{|\vec{b}|}$

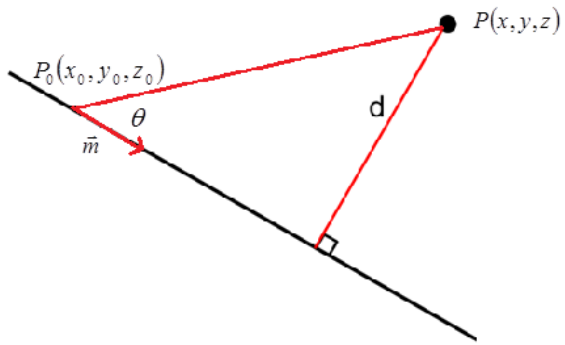
To determine the minimum distance from a point $P(x_1, y_1)$ to the line $Ax + By + C = 0$, let $P_0(x_0, y_0)$ be a known point on the line and let d be the minimum distance.



Example 1: Determine the minimum distance from (6,-1) to the line $4x + y - 10 = 0$

Minimum distance in R^3

Recall: Lines in R^3 do not have unique normal vectors.



Example 2: Determine the minimum distance from (3,-2,1) to the line $\vec{r} = (1,0,-5) + t(2,-2,3)$, $t \in \Re$.